Beam mold: drilling templates

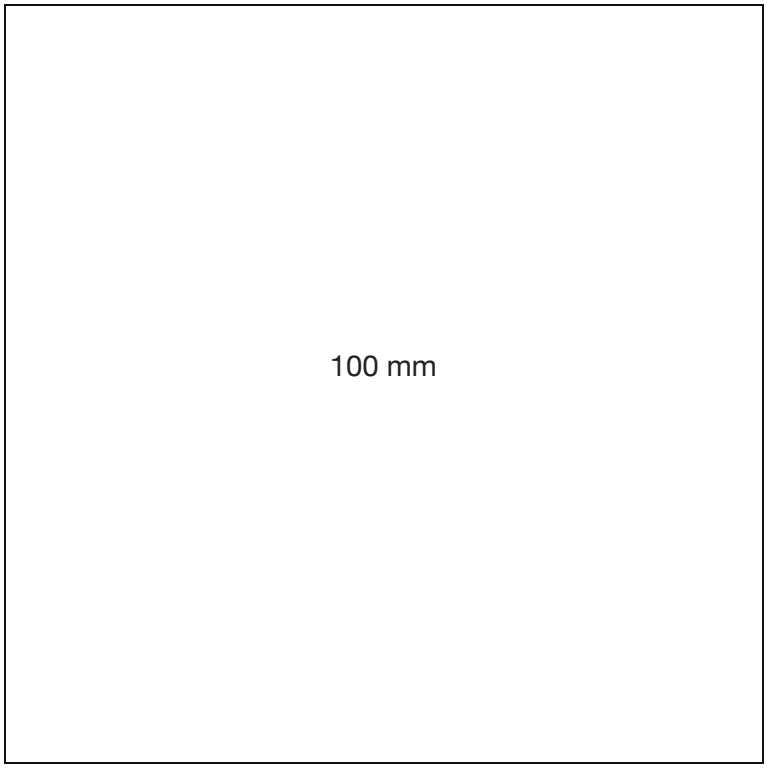
DIY CNC epoxy-granite mill · October 2026 · full size, US Letter

Printing

1. Print at **Actual size / 100 %**. Turn off "Fit to page" and "Scale to fit". Letter paper, portrait.
2. Measure the square below: it must be 100 mm (and the bar on every page 50 mm). If it is off by more than 1 mm, fix the scaling and print again.
3. Only pages with holes are printed. Every page has a grid labeled in the board's own millimeters, from the bottom-left corner as you look at the marked face, so you can place any page on its own.

Marking a board

1. Lay the board on the bench with the face named at the top of its pages up (the cast face, or the top face for bridges and cleats).
2. Start with the page that has the board's corner on it: line its printed edges up with the board's edges and tape it down. Add the next pages overlapping by about 12 mm, with the grid lines on top of each other; trim one edge if it helps. Where a page shows no board edge, measure to a labeled grid line with a square instead.
3. Push an awl or a center punch through every crosshair. The circle is the finished hole size, the number is the hole's number in the guide's table, and the color is its size: the legend at the top of each page gives the drill.
4. Pull the paper off, and drill from the marked face (clamp a scrap underneath). Drill a 1/8 in pilot first for anything 13 mm or bigger; the 63.5 mm holes are a 2-1/2 in hole saw.
5. Walls: pencil the dashed **pour line** (25 mm below the top edge) on the cast face, or mark it later with the printed pour-line gauge. The casting is filled to this line.
6. Boards marked "x2" are the same board twice: mark both from one print.



Pages

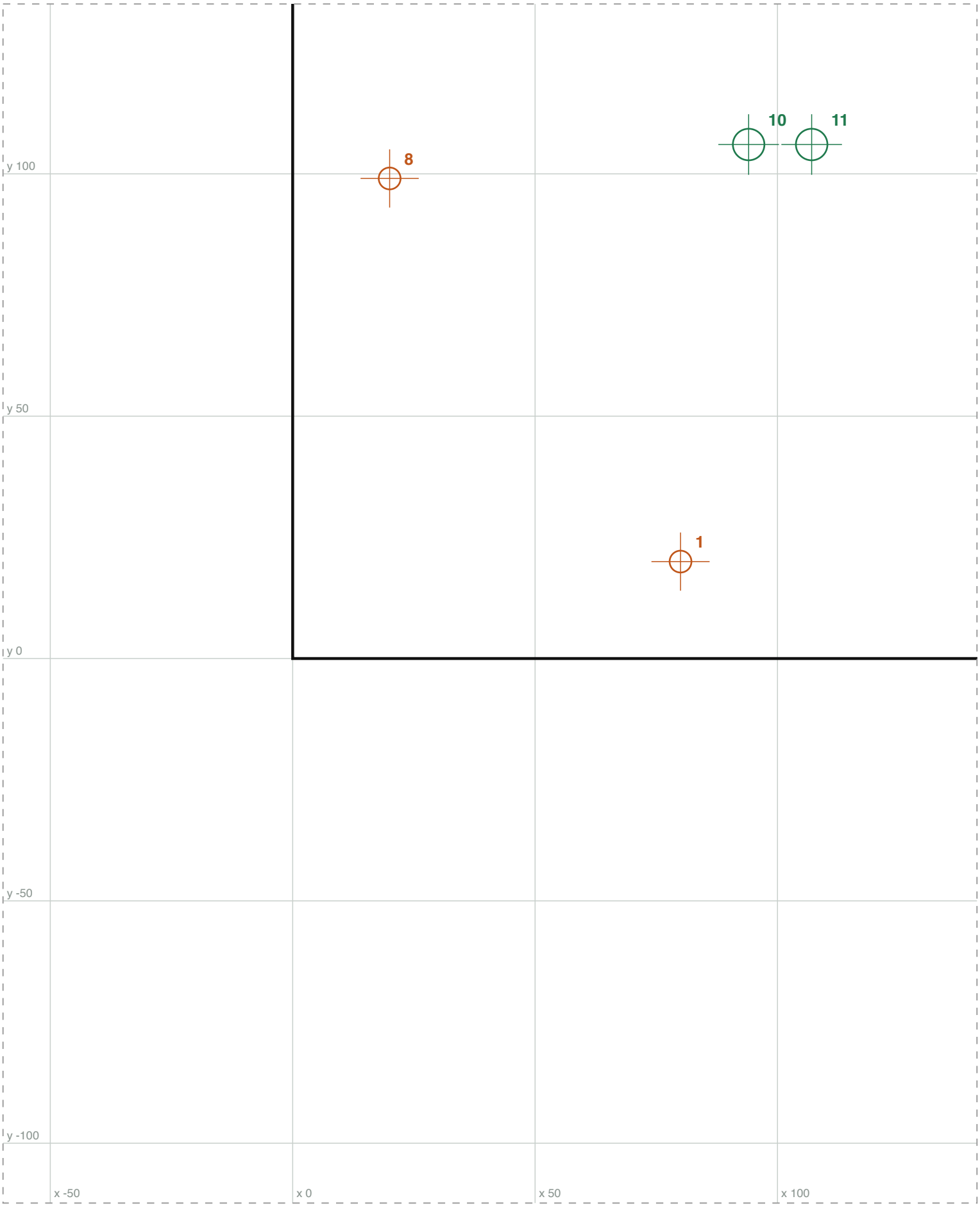
Board	Qty	Pages	Holes
Beam floor	1	2-9 (8)	24
Beam top wall	1	10-12 (3)	14
Beam bottom wall	1	13-14 (2)	12
Beam end wall, left	1	15-15 (1)	4

Beam floor

page B1 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

4.5 mm (11/64 in) 6.5 mm (17/64 in)

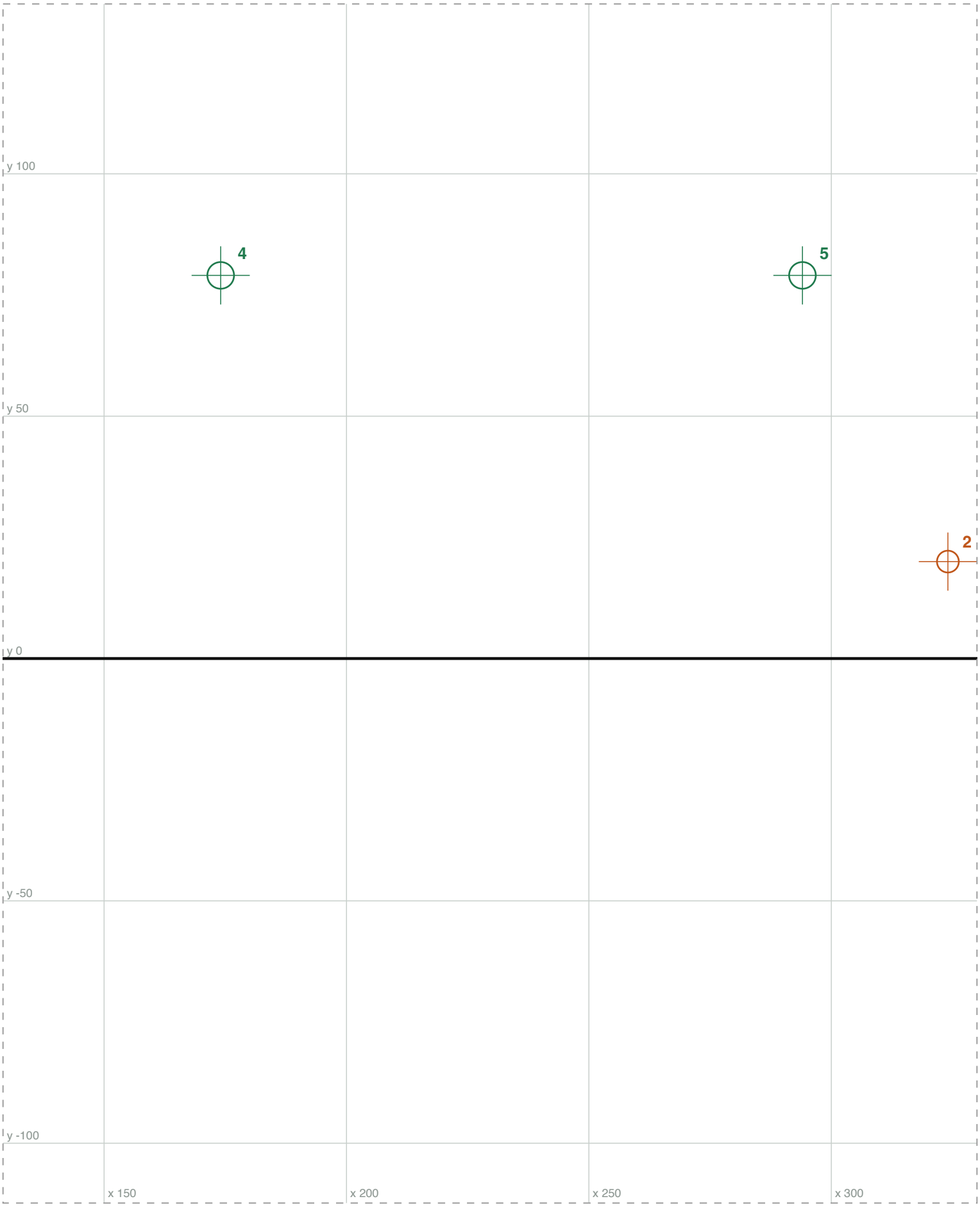


Beam floor

page B2 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

● 4.5 mm (11/64 in) ● 5.5 mm (7/32 in)

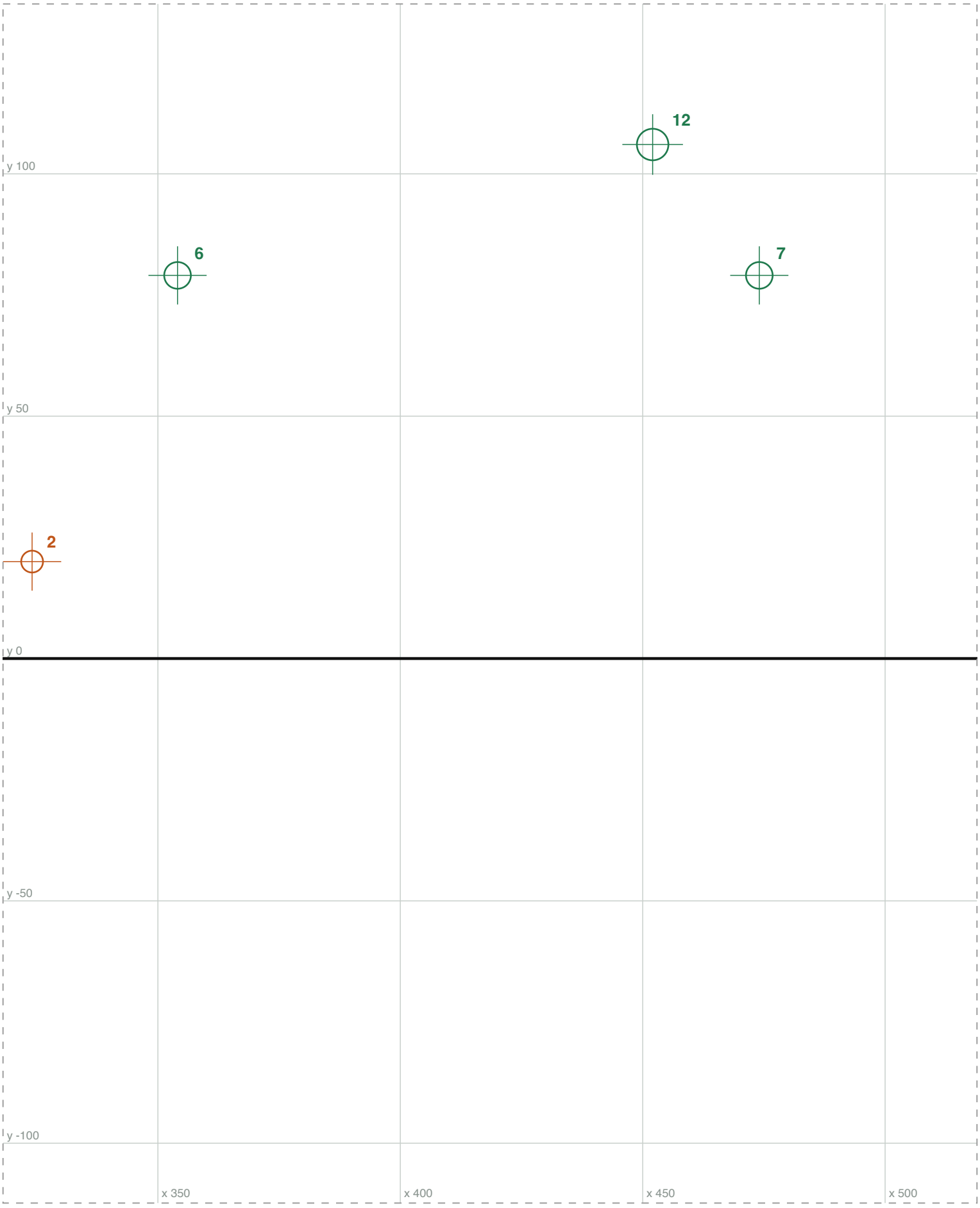


Beam floor

page B3 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

4.5 mm (11/64 in) · 5.5 mm (7/32 in) · 6.5 mm (17/64 in)

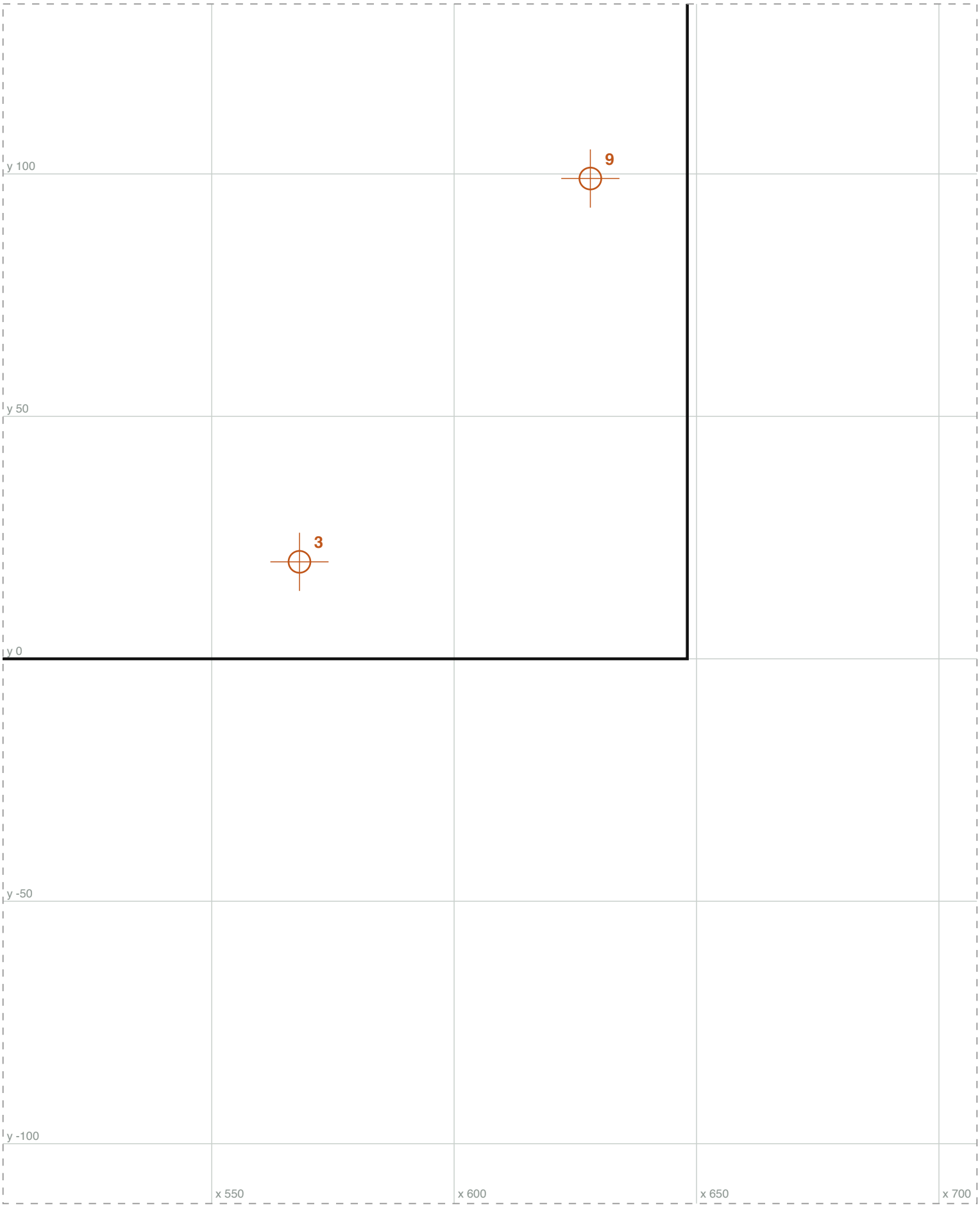


Beam floor

page B4 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

4.5 mm (11/64 in)

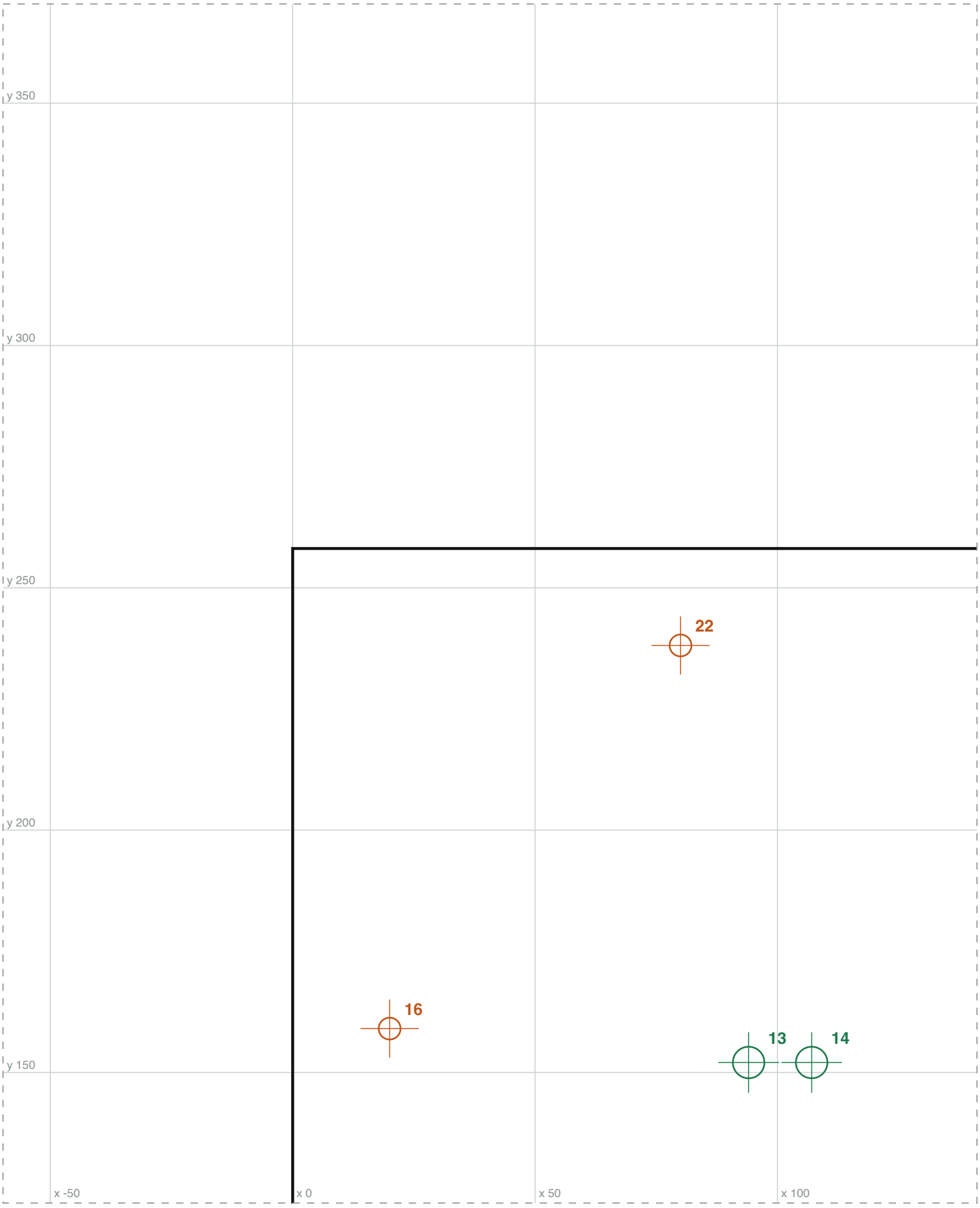


Beam floor

page A1 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

4.5 mm (11/64 in) 6.5 mm (17/64 in)

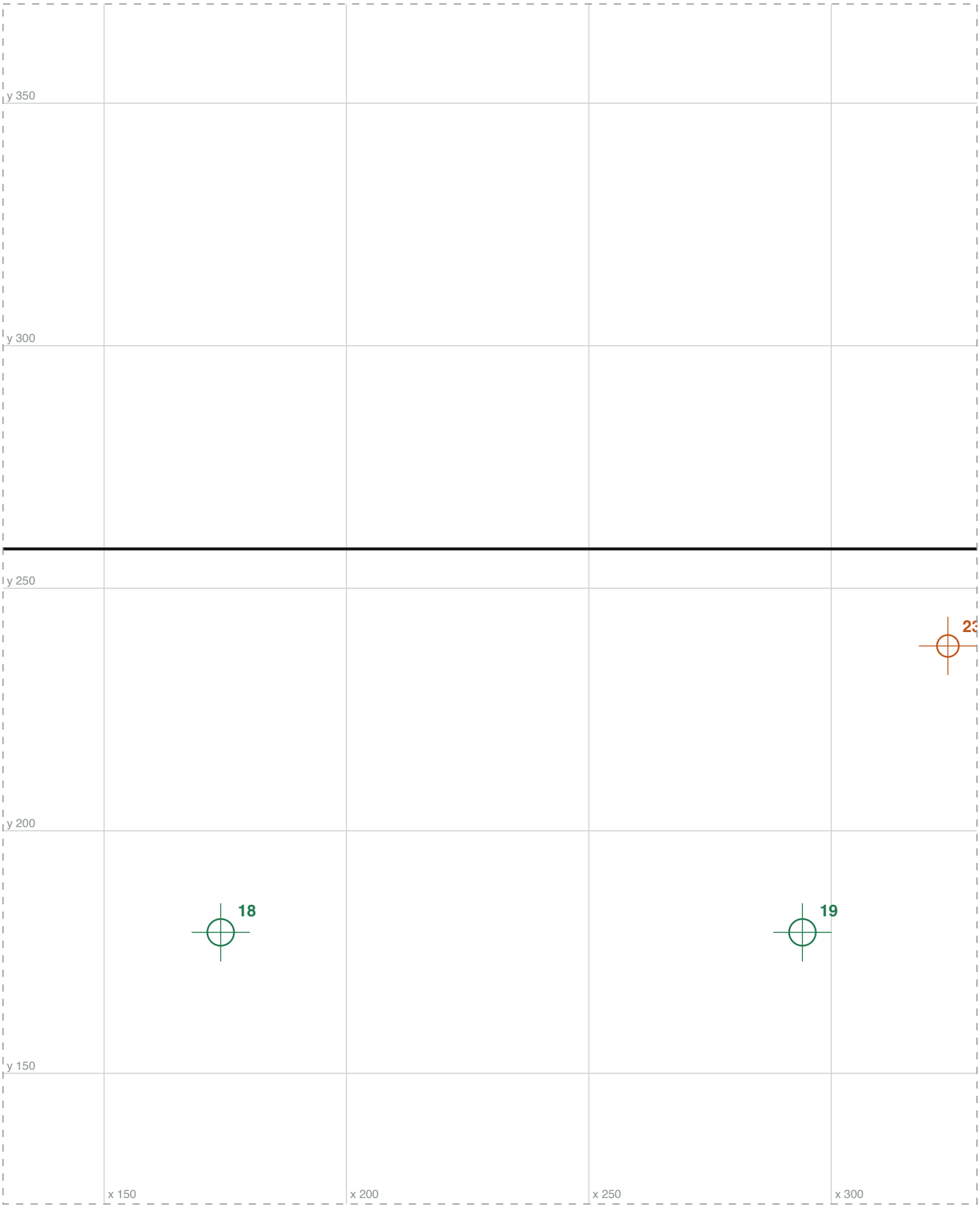


Beam floor

page A2 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

● 4.5 mm (11/64 in) ● 5.5 mm (7/32 in)

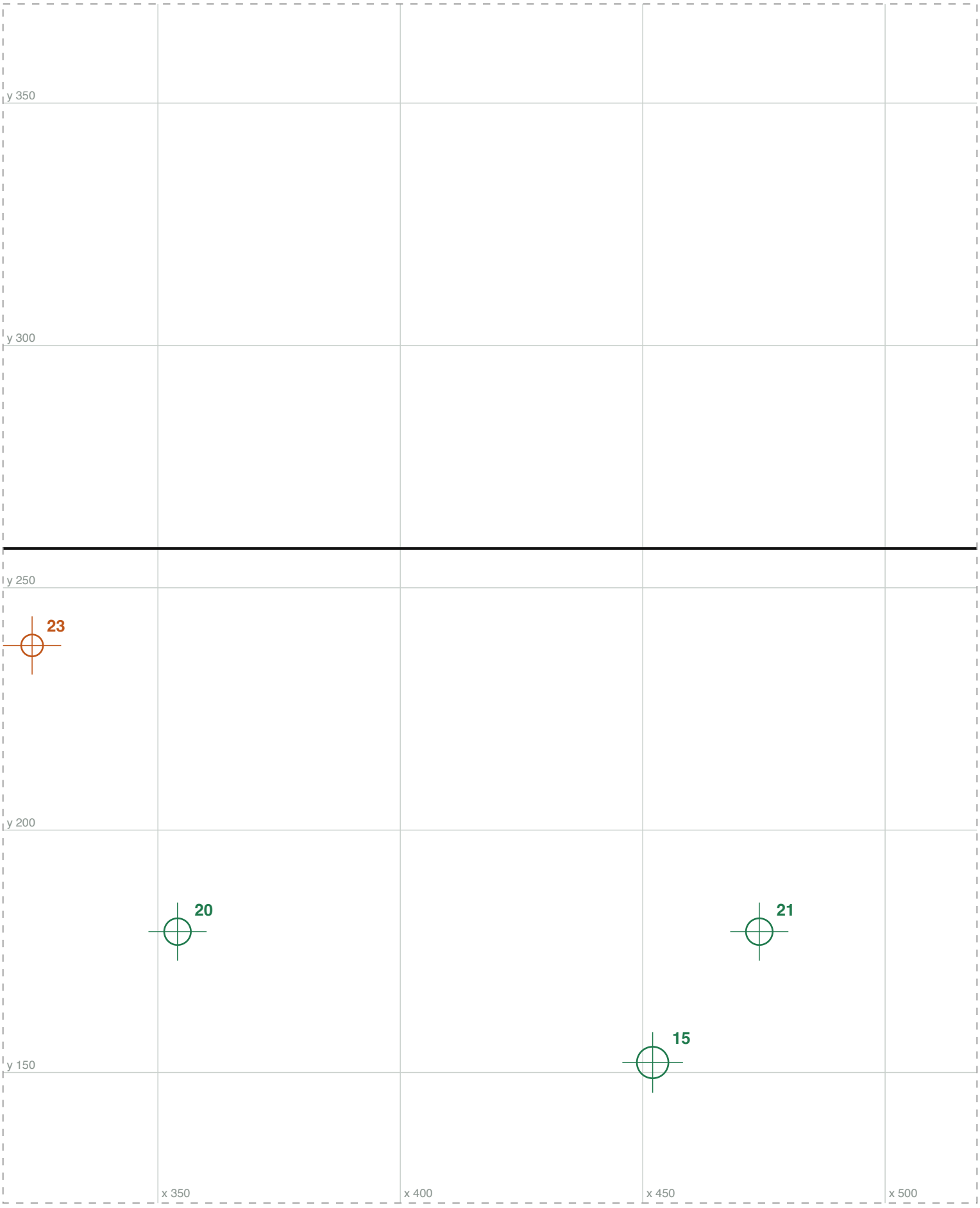


Beam floor

page A3 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

● 4.5 mm (11/64 in) ● 5.5 mm (7/32 in) ● 6.5 mm (17/64 in)

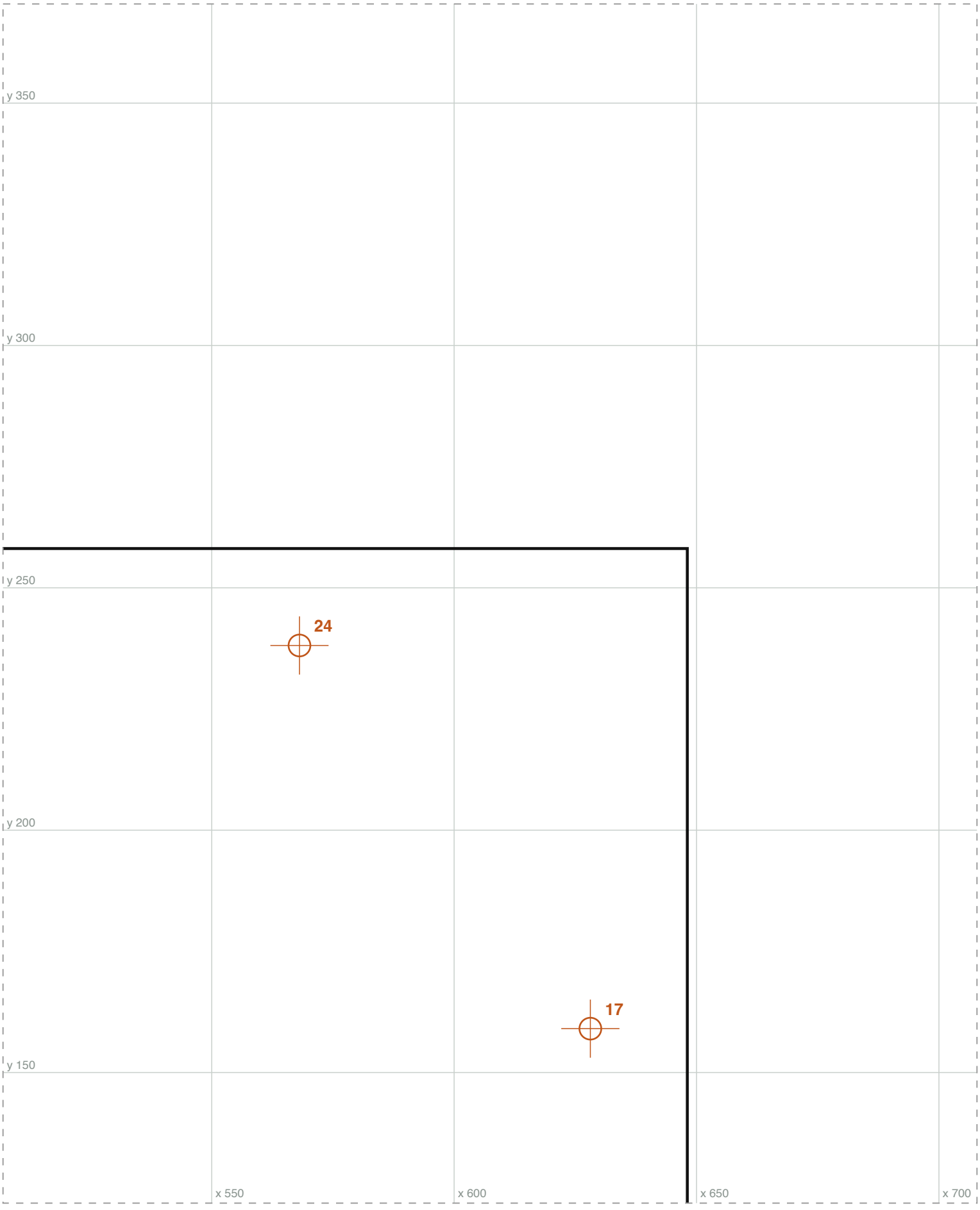


Beam floor

page A4 · full size · lay it on the cast face (the side the mix touches) · board 648 x 258 mm

50 mm: measure it

4.5 mm (11/64 in)

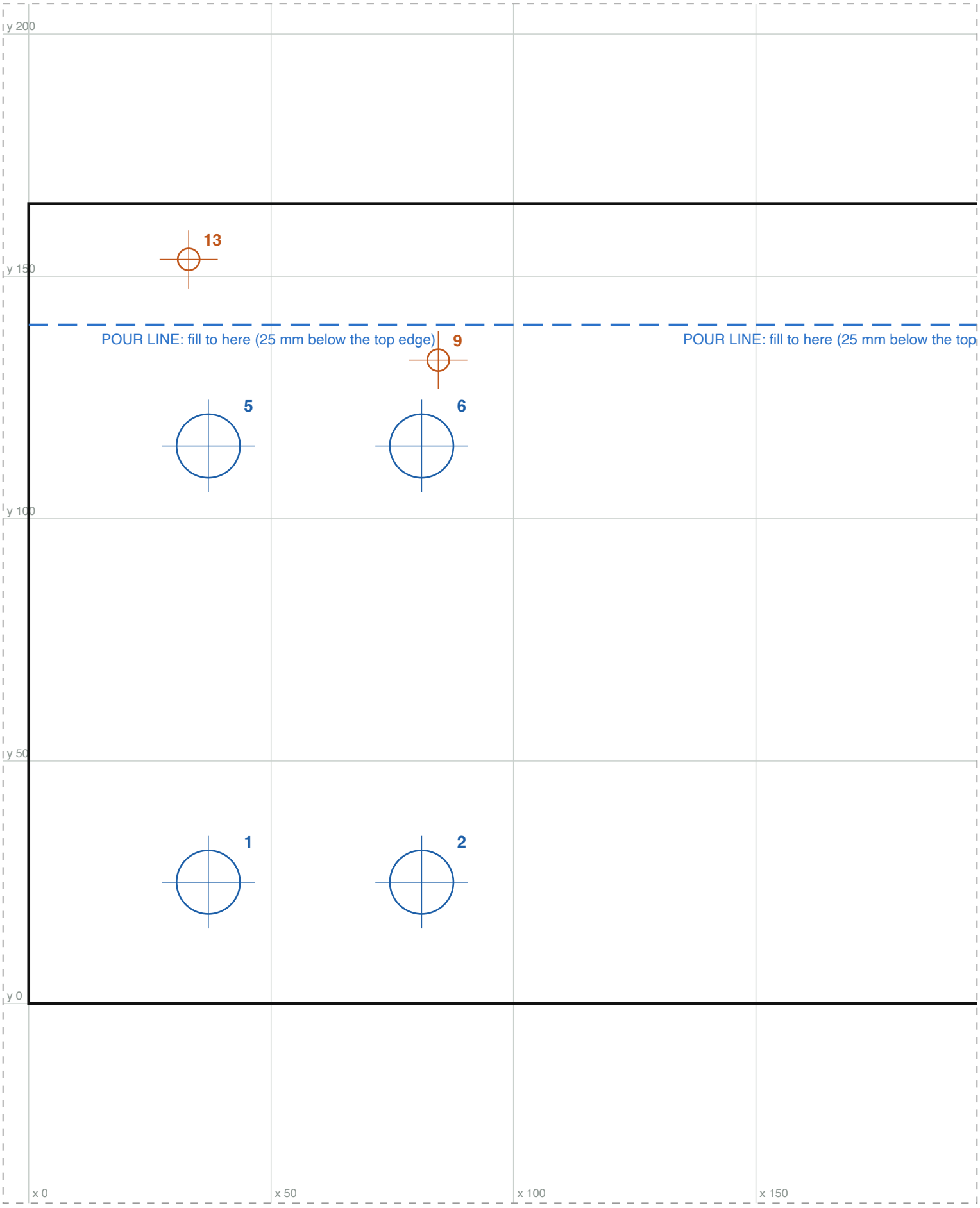


Beam top wall

page A1 · full size · lay it on the cast face (the side the mix touches) · board 568 x 165 mm

50 mm: measure it

4.5 mm (11/64 in) 13.1 mm (33/64 in)

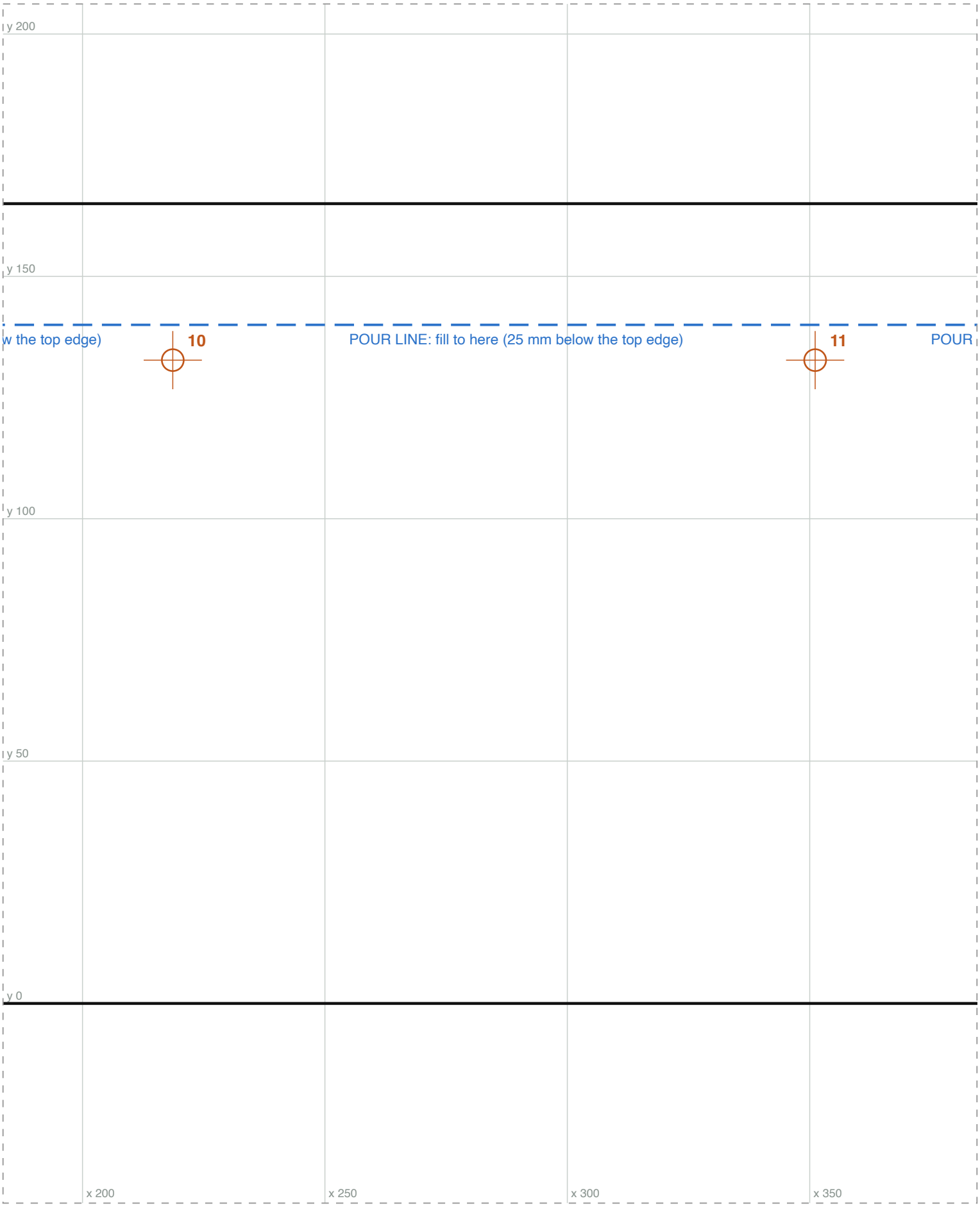


Beam top wall

page A2 · full size · lay it on the cast face (the side the mix touches) · board 568 x 165 mm

50 mm: measure it

4.5 mm (11/64 in)

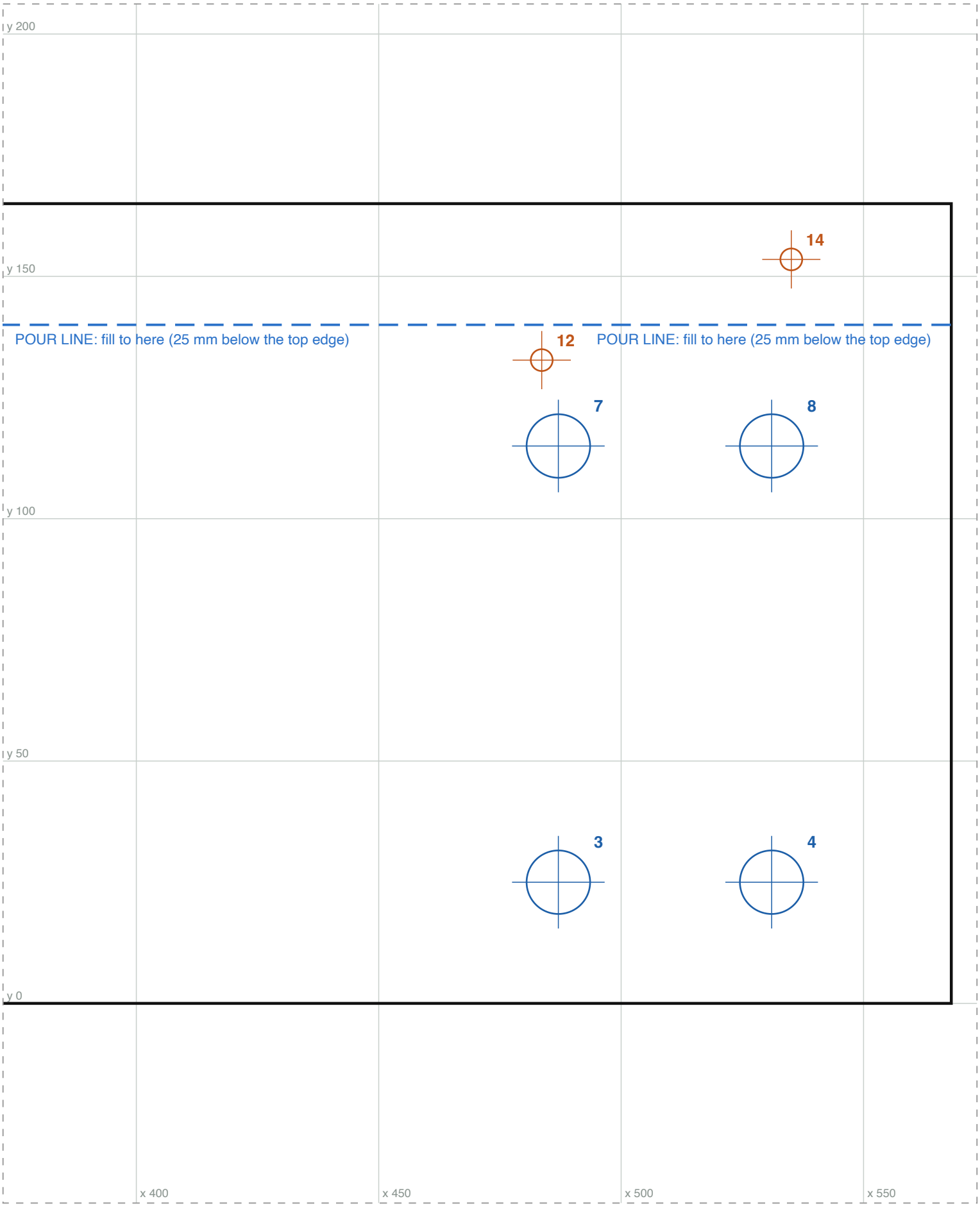


Beam top wall

page A3 · full size · lay it on the cast face (the side the mix touches) · board 568 x 165 mm

50 mm: measure it

4.5 mm (11/64 in) 13.1 mm (33/64 in)

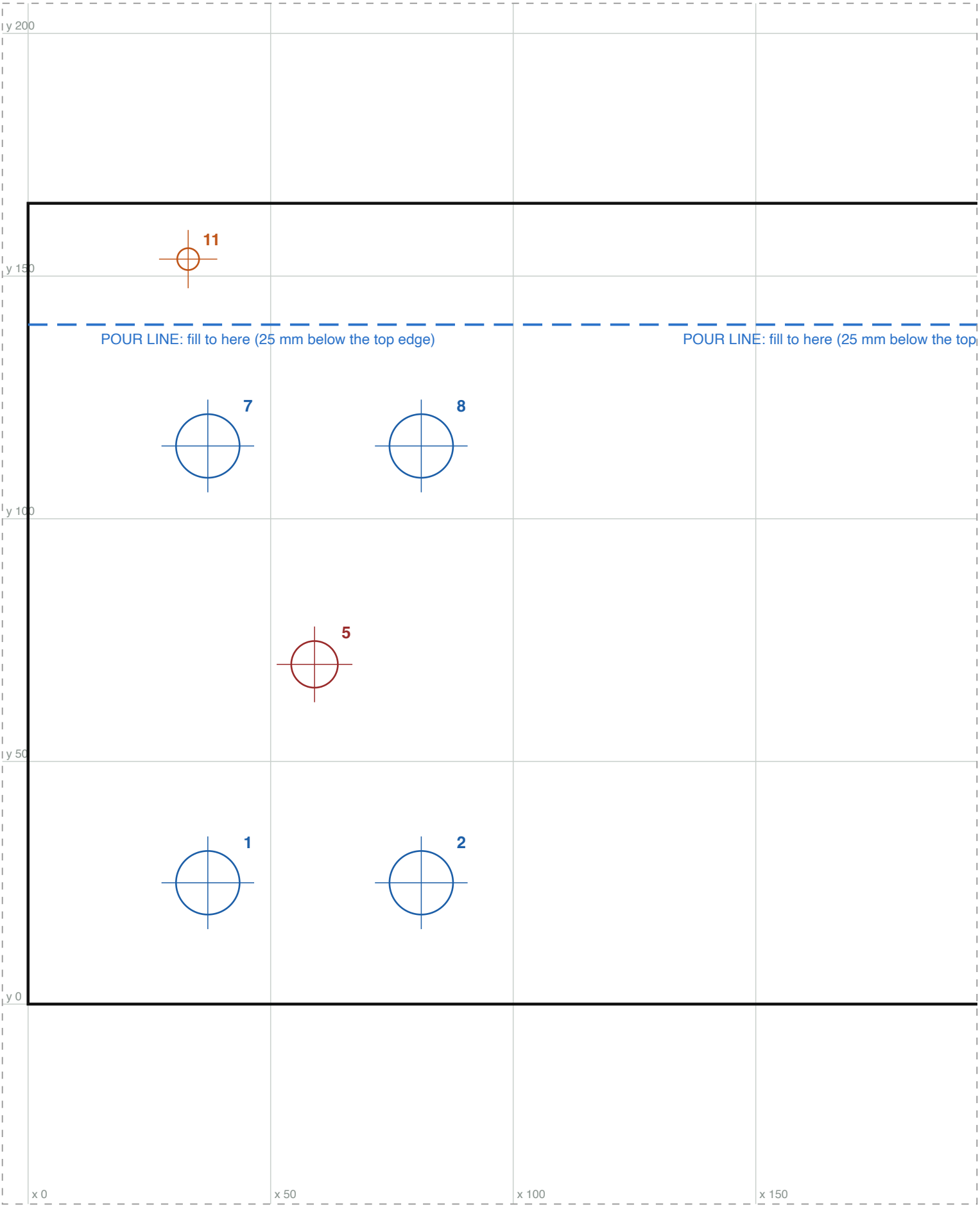


Beam bottom wall

page A1 · full size · lay it on the cast face (the side the mix touches) · board 568 x 165 mm

50 mm: measure it

4.5 mm (11/64 in) 9.6 mm (25/64 in) 13.1 mm (33/64 in)

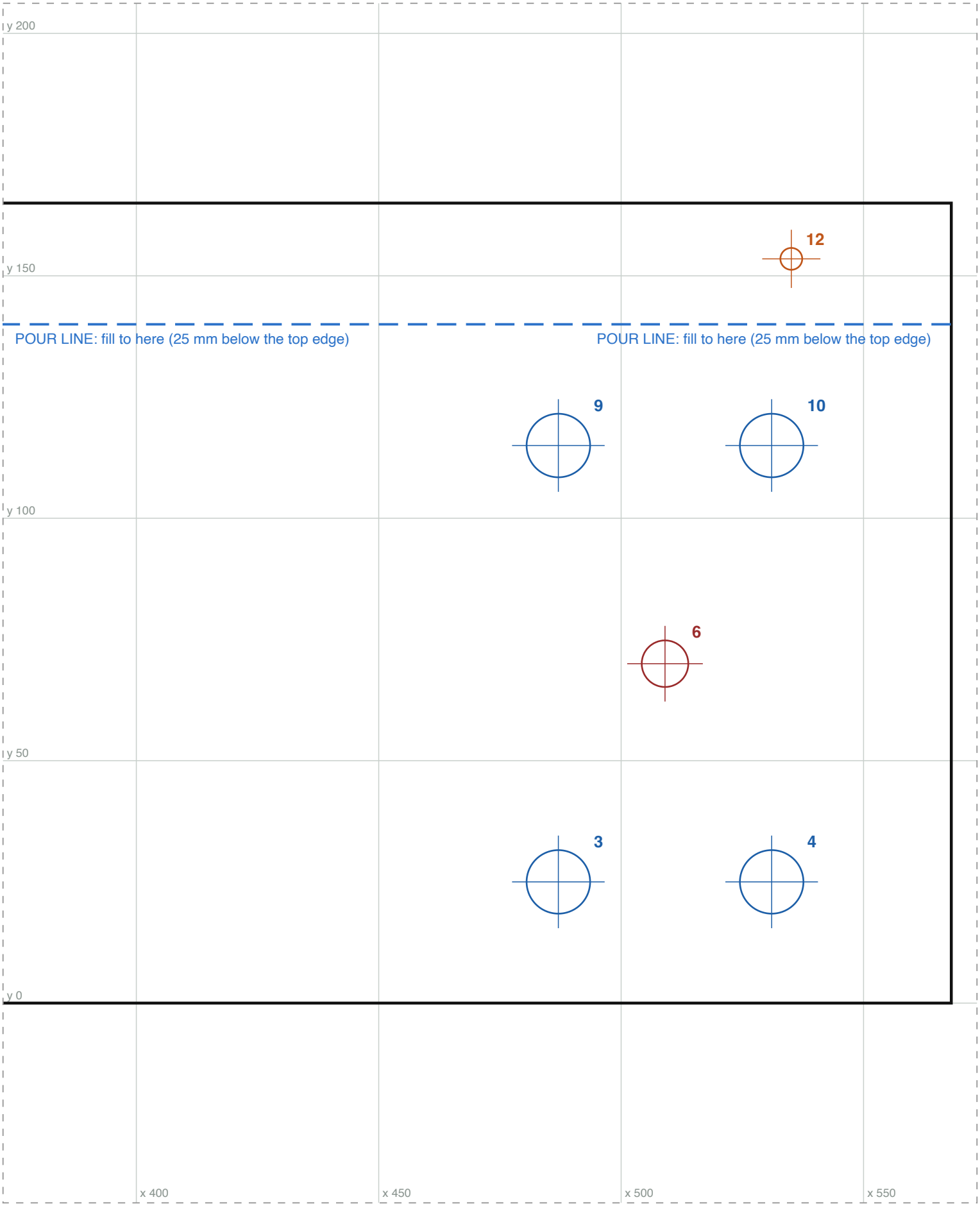


Beam bottom wall

page A3 · full size · lay it on the cast face (the side the mix touches) · board 568 x 165 mm

50 mm: measure it

4.5 mm (11/64 in) 9.6 mm (25/64 in) 13.1 mm (33/64 in)

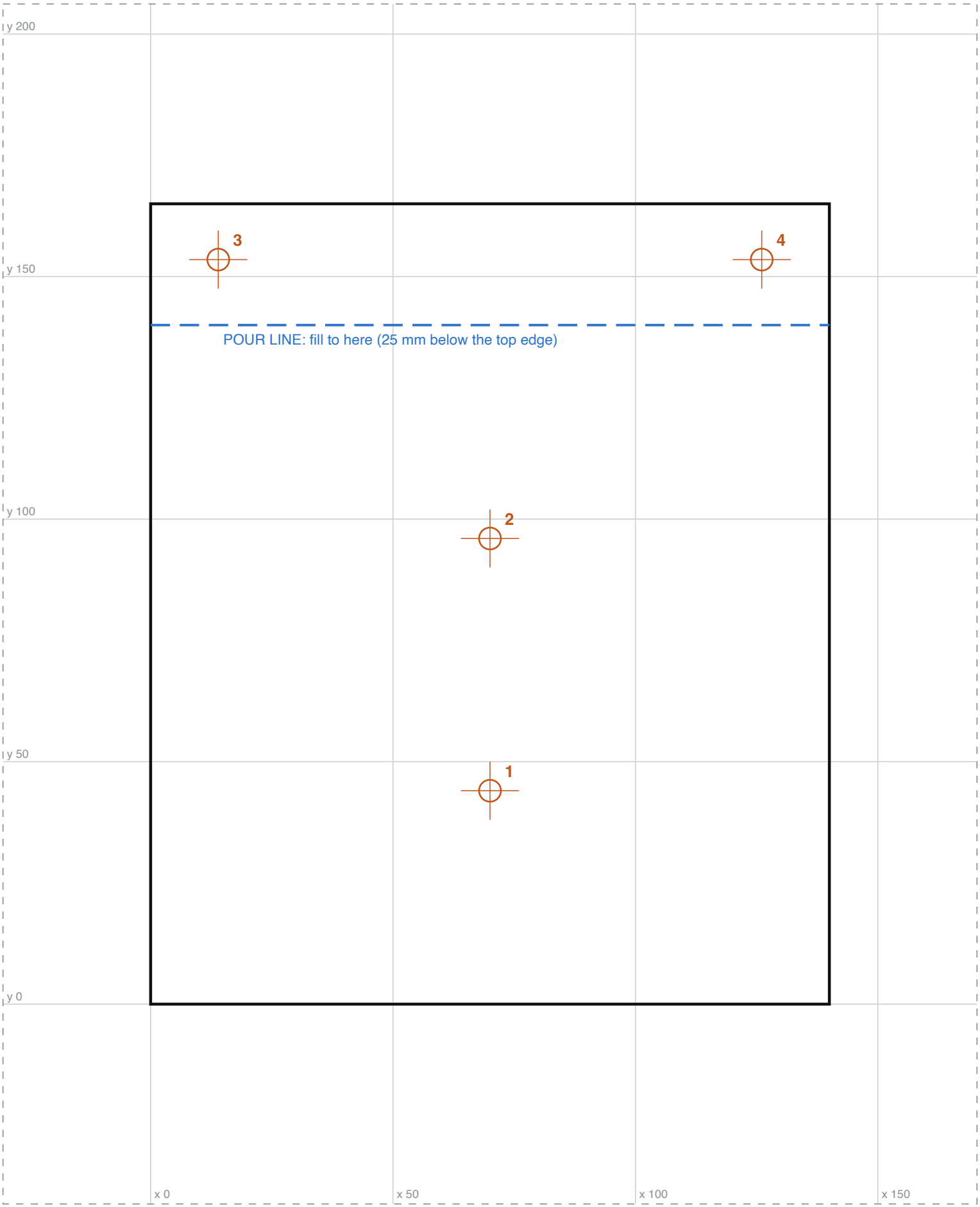


Beam end wall, left

page A1 · full size · lay it on the cast face (the side the mix touches) · board 140 x 165 mm

50 mm: measure it

● 4.5 mm (11/64 in)

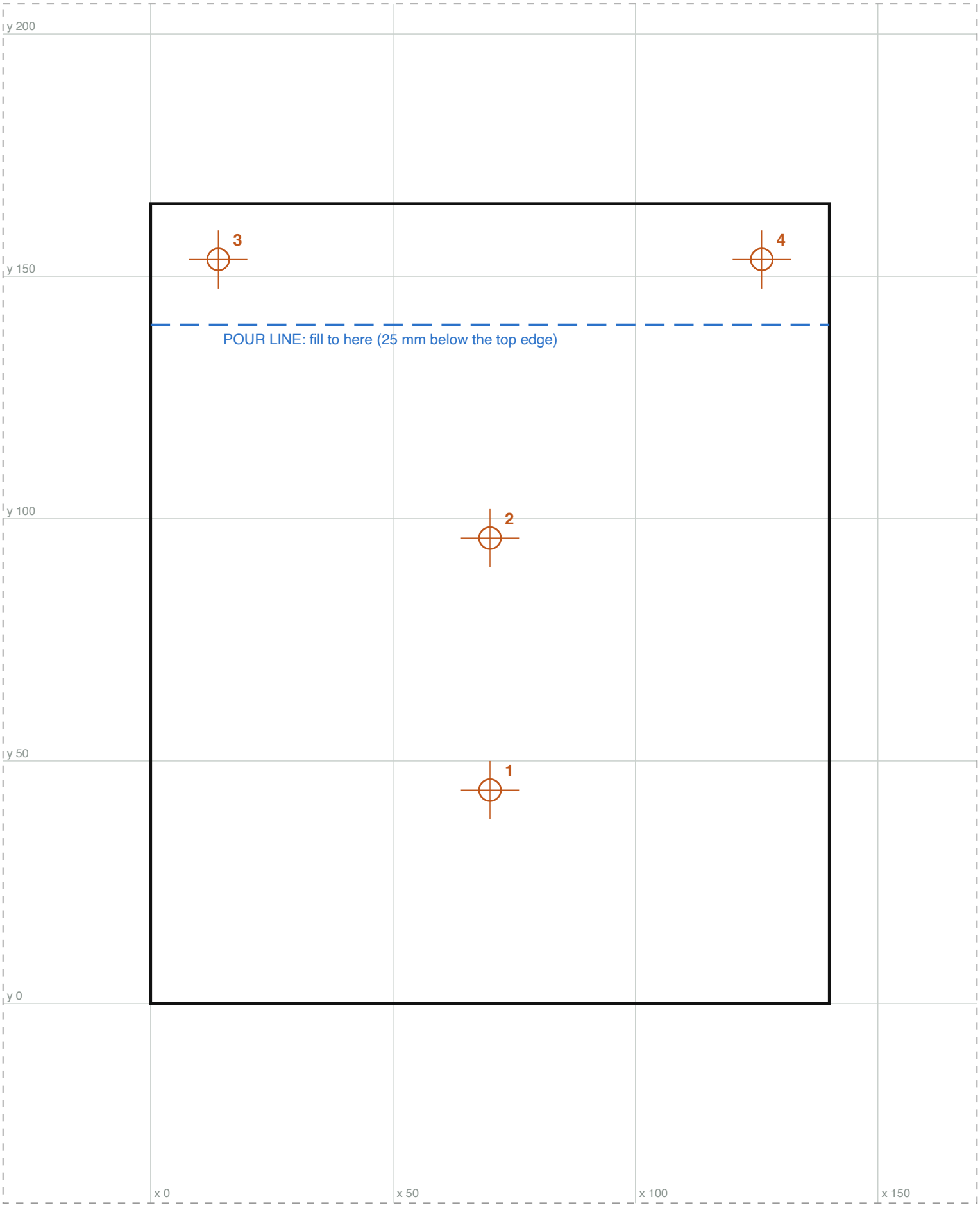


Beam end wall, right

page A1 · full size · lay it on the cast face (the side the mix touches) · board 140 x 165 mm

50 mm: measure it

4.5 mm (11/64 in)

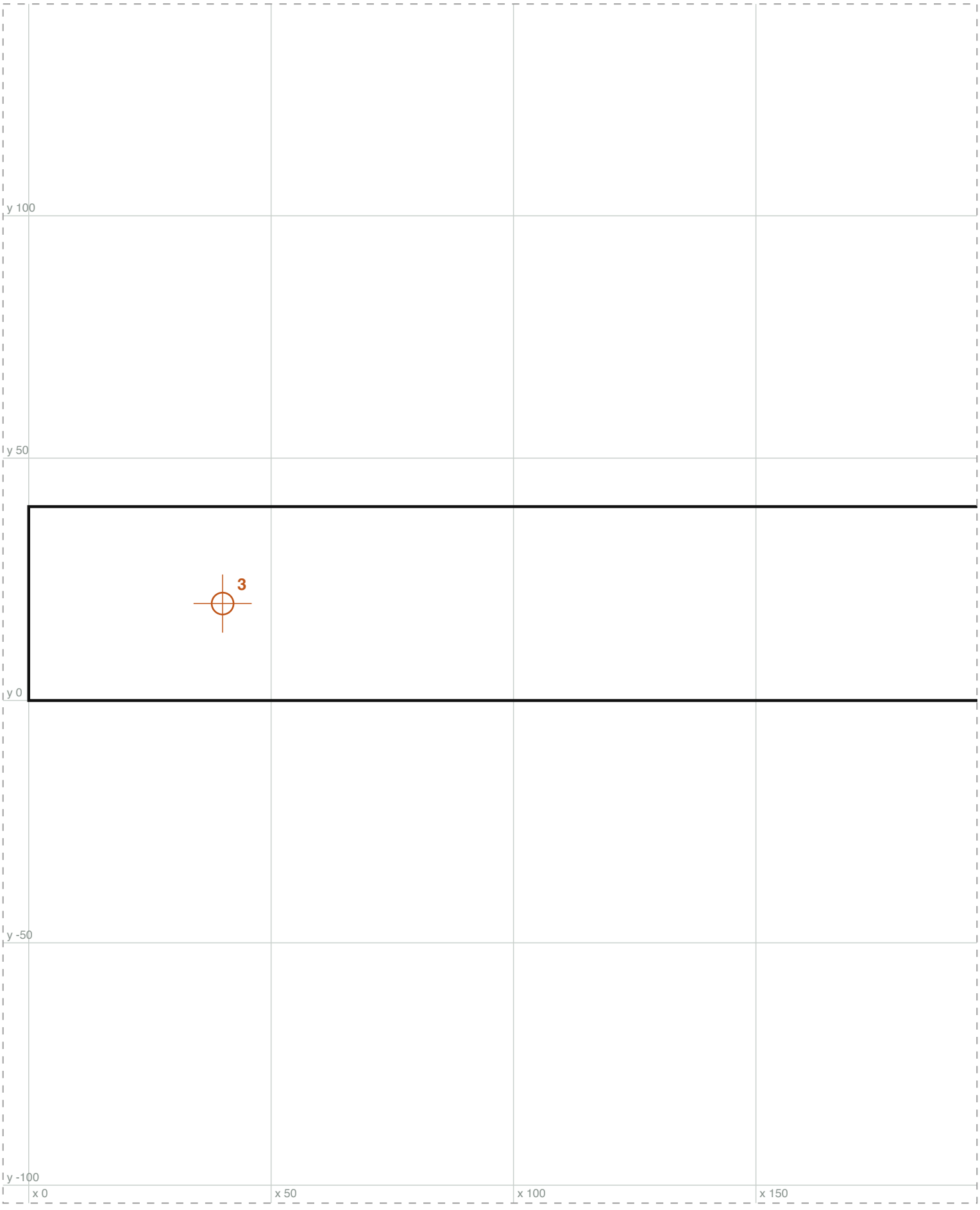


Beam cleat, top

page A1 · full size · lay it on the top (outside) face · board 568 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)

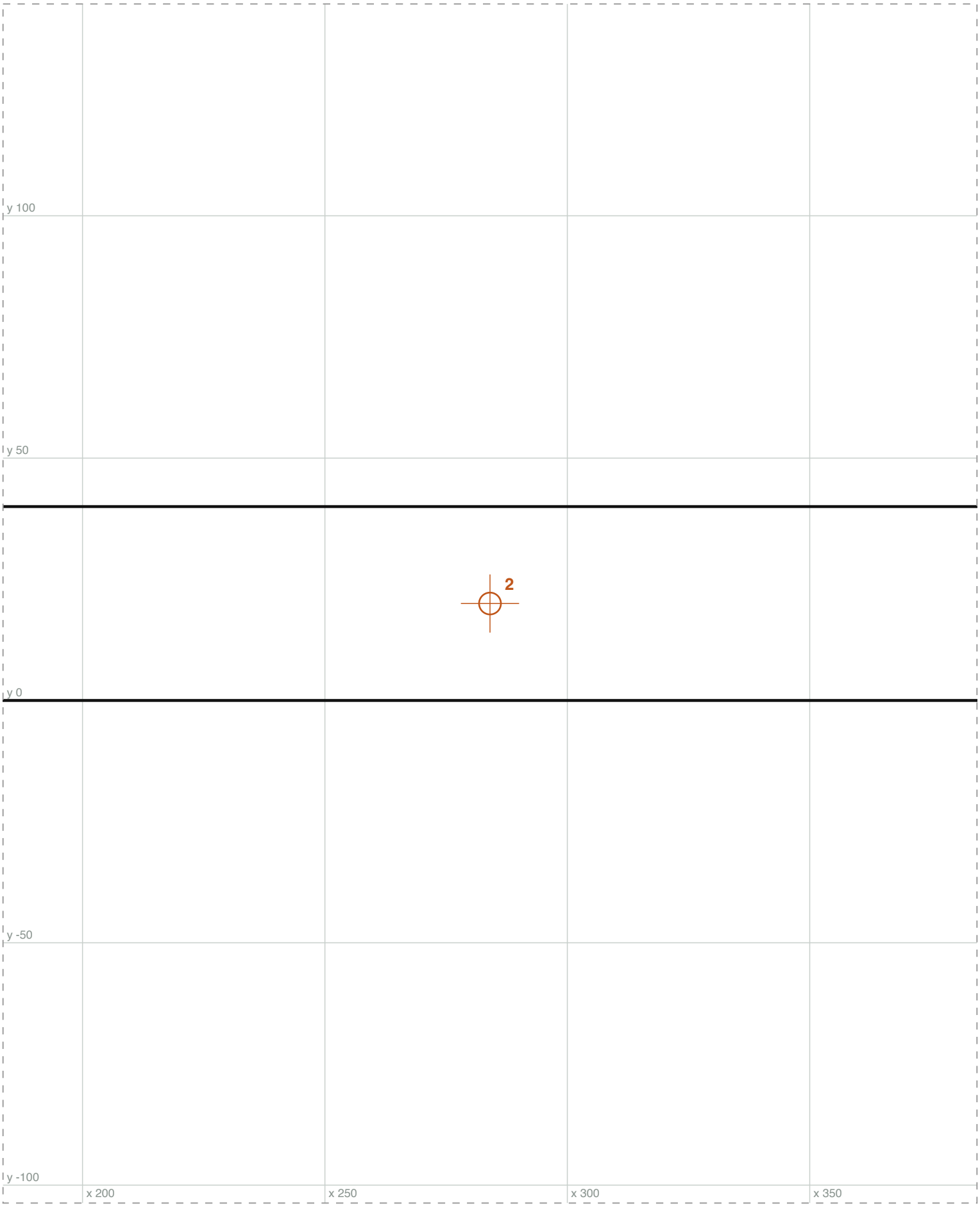


Beam cleat, top

page A2 · full size · lay it on the top (outside) face · board 568 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)

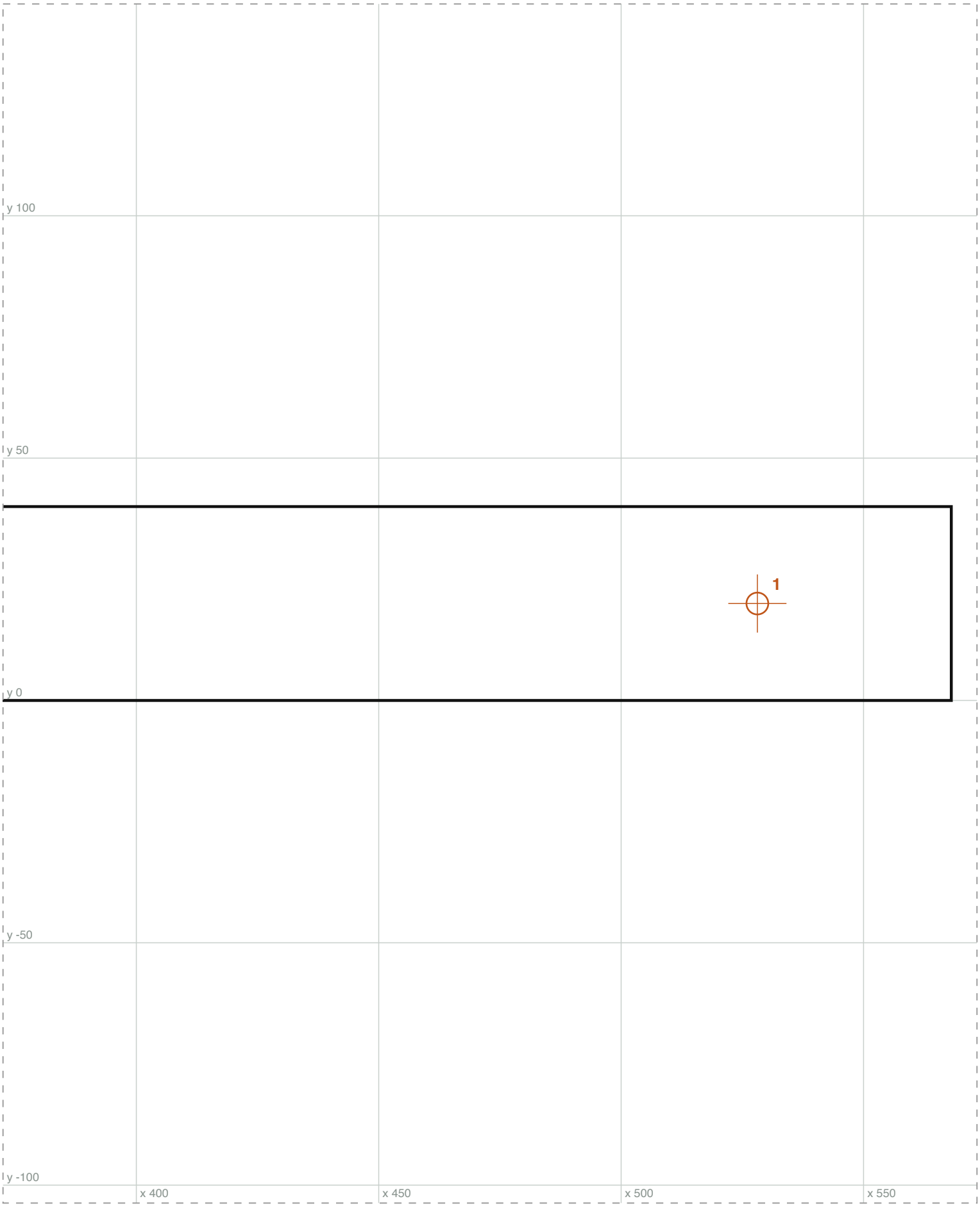


Beam cleat, top

page A3 · full size · lay it on the top (outside) face · board 568 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)

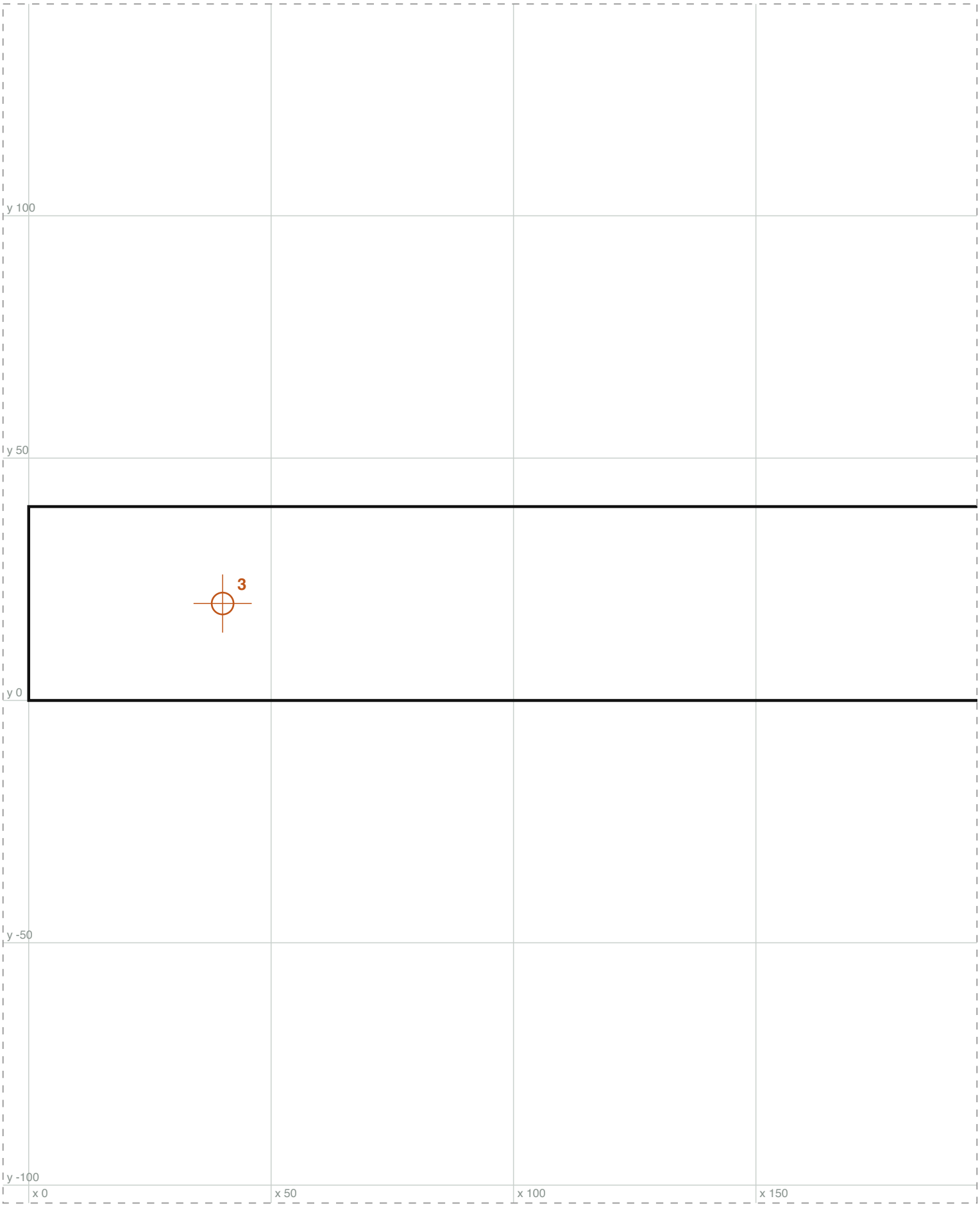


Beam cleat, bottom

page A1 · full size · lay it on the top (outside) face · board 568 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)

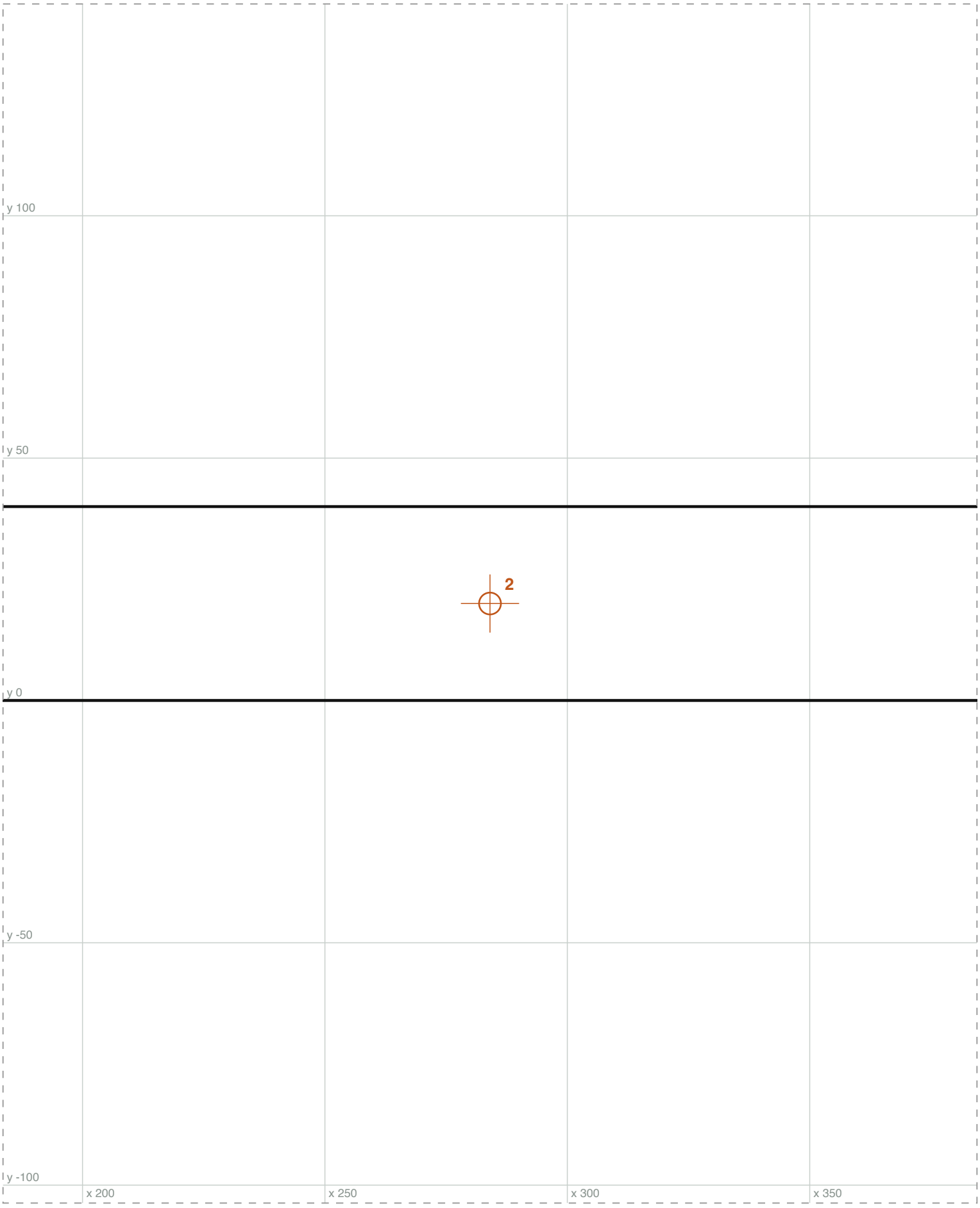


Beam cleat, bottom

page A2 · full size · lay it on the top (outside) face · board 568 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)

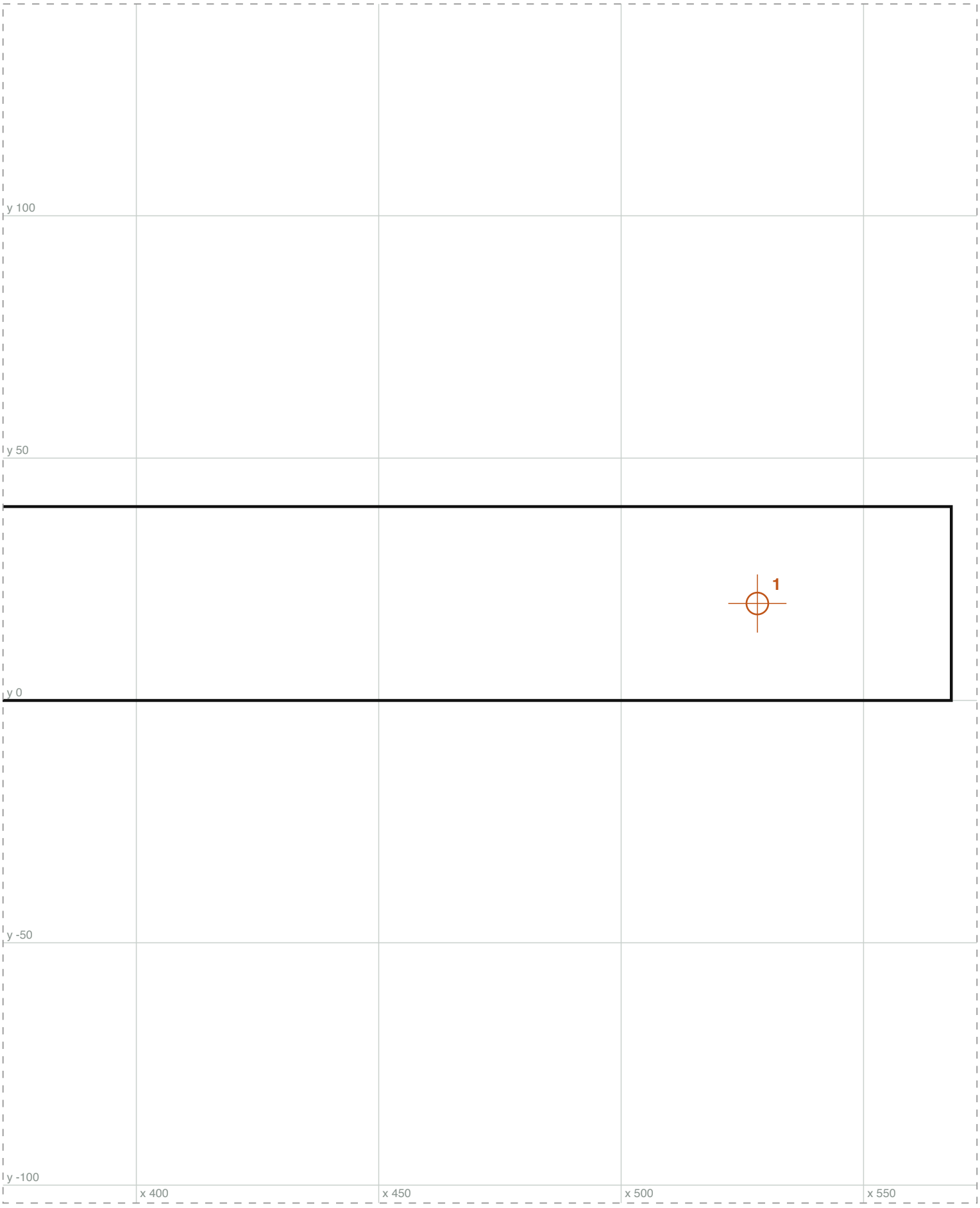


Beam cleat, bottom

page A3 · full size · lay it on the top (outside) face · board 568 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)

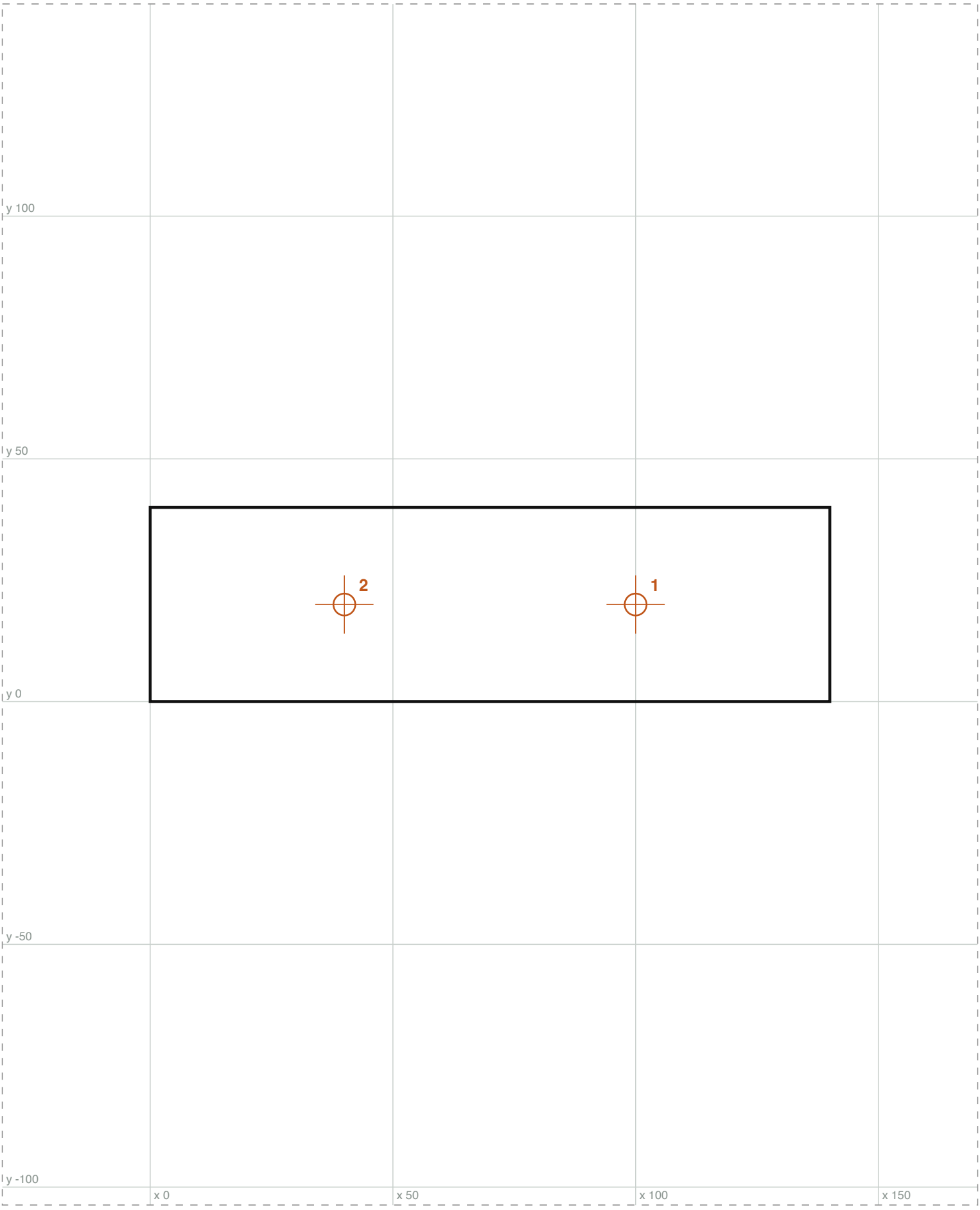


Beam cleat, left

page A1 · full size · lay it on the top (outside) face · board 140 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)



Beam cleat, right

page A1 · full size · lay it on the top (outside) face · board 140 x 40 mm

50 mm: measure it

● 4.5 mm (11/64 in)

